

三. 爱因斯坦方程的解

1. 稳态时空, 静态时空

(M, g_{ab}) 稳态

Stationary

引力场不随时间变

若存在类时 Killing 场 (g_{ab} 则为稳态度规)

取 $t=x^0$, 其余 $\{x^i\}$ 任取: $g_{\mu\nu}$ 为分量, $\partial g_{\mu\nu}/\partial t = (\mathcal{L}_\xi g)_{\mu\nu} = 0 \rightarrow$ 稳态

若 $\exists \{x^i\}$ 使 $\partial g_{\mu\nu}/\partial x^0 = 0 \rightarrow$ Killing 场 $\xi^a = (\partial/\partial x^0)^a \rightarrow$ 稳态

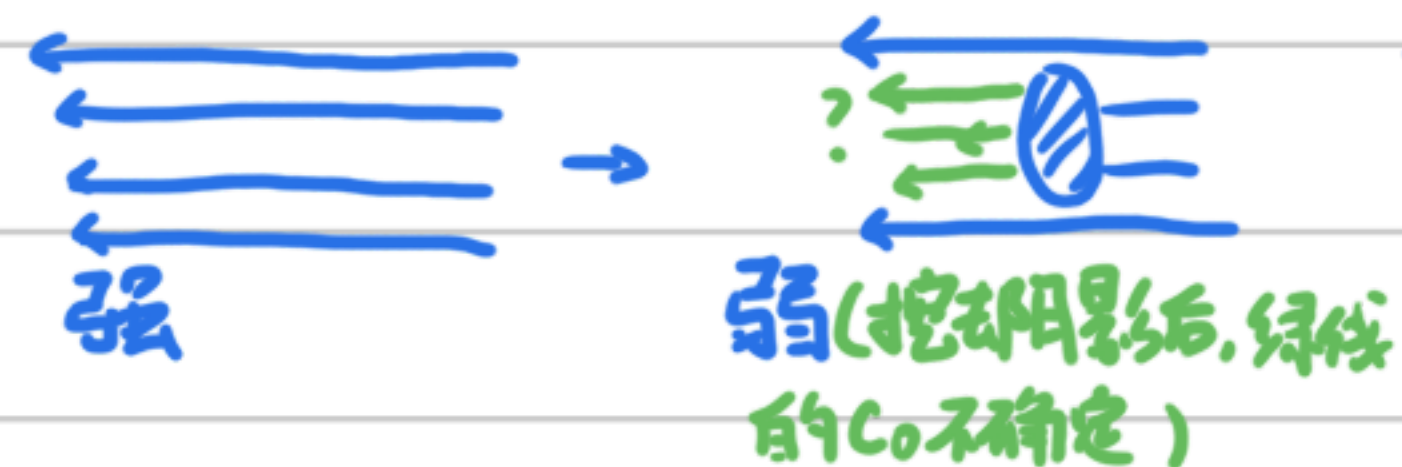
Killing 场定义

这里“ t ”为坐标时间, 与观者固有时“ τ ”不同,
 t 表示一种对称性, 不代表时间流动速率
 $d\tau = \sqrt{-g_{00}} dt$

强弱 Killing 场:

弱: $\mathcal{L}_\xi g_{ab} = 0$ 或 $\nabla_a \xi_b = 0$ 即 (局部)

强: ξ^a 完备 (t 取值全 $\mathbb{R}$) (全局)



时轴正交坐标系

time-orthogonal coordinate system

归一化: $d\tau = \sqrt{-g_{00}} dt$

静态: $\exists$ Killing 类时场 ξ^a , $\exists$ 超曲面 Σ , Σ 与 ξ^a 处处正交

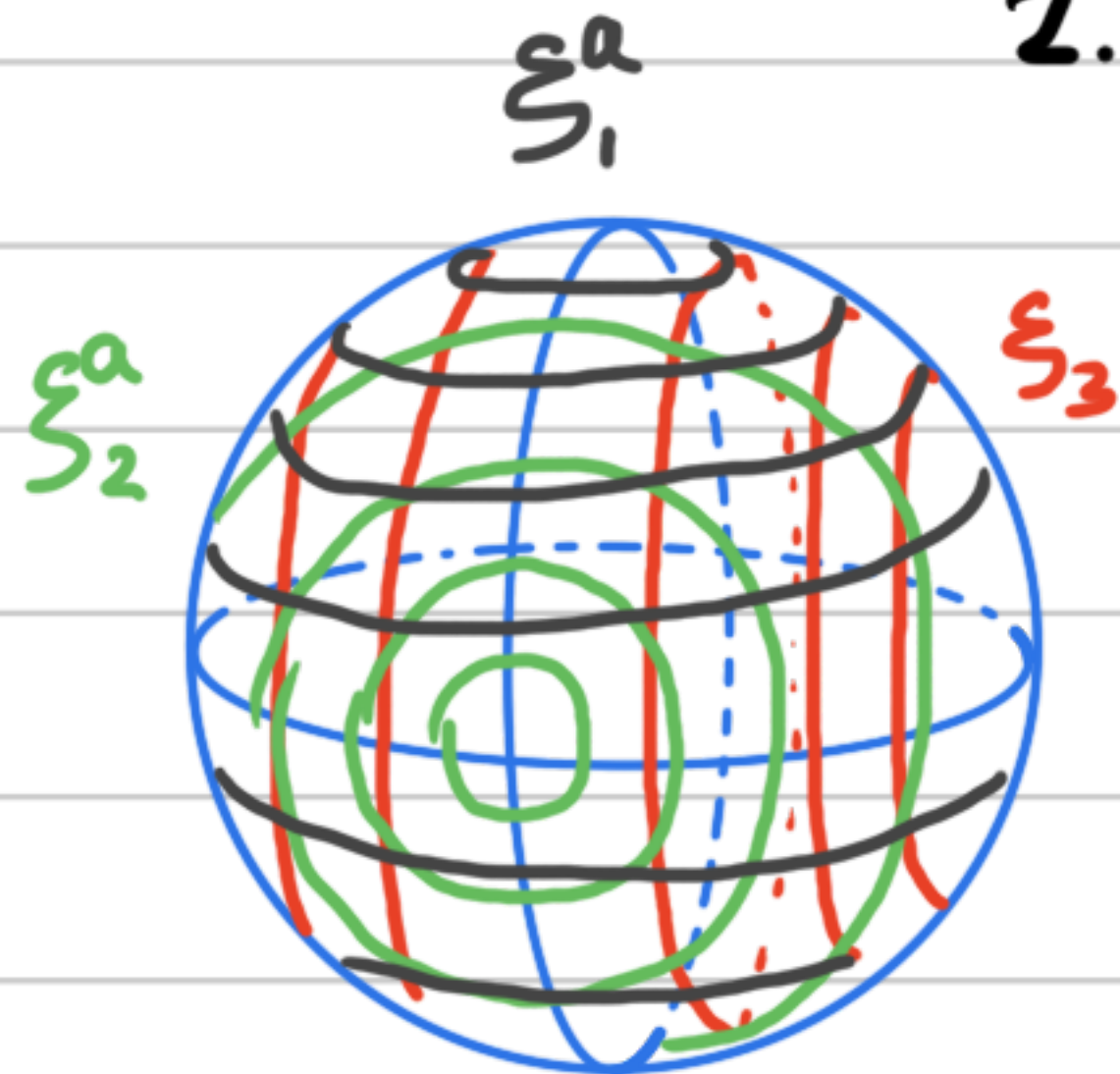
static
时间平移不变性

Claim: 若 $\Sigma_0 = \{p \in M | t(p) = 0\}$ 与 ξ^a 正交, 则 $\Sigma_t = \{p \in M | t(p) = t\}$ 也正交

Pf: $Z_t = \phi_t[\Sigma_0]$, ξ^a 有: $\mathcal{L}_\xi \xi^a = 0 \rightarrow \phi_t^* \xi^a = \xi^a \rightarrow (\partial/\partial t)^a$ 与 ξ^a 切关系不变

取 $\xi^a = (\partial/\partial t)^a$, $g_{\mu\nu}$ 有: $g_{0i} = g_{ab} (\partial/\partial t)^a (\partial/\partial x^i)^b = 0 \rightarrow ds^2 = g_{00}(x^i) dt^2 + g_{ij}(x^i) dx^i dx^j$
 正交 $\partial g_{\mu\nu}/\partial t = 0$
 均时空分离, 可沿用空间对称 $\{x^i\}$

2. 球对称时空



球度规 (i,j,k) $ds^2 = r^2(d\theta^2 + \sin^2\theta d\varphi^2) + dr^2$

球面 $[0, \varphi]$ $\xi_1^a = (\partial/\partial \varphi)^a$ 为 Killing 场

$\xi_2^a = (\partial/\partial \theta)^a \cdot \sin\varphi + (\partial/\partial \varphi)^a \cot\theta \cos\varphi$

$\xi_3^a = [\xi_1, \xi_2]^a = (\partial/\partial \theta)^a \cos\varphi - (\partial/\partial \varphi)^a \cot\theta \sin\varphi$

$\xi_1, \xi_2 \rightarrow \hat{g}_{ab}$ 在 $(\partial/\partial \theta)_a, (\partial/\partial \varphi)_b$ 坐标下 $\hat{g}_1 = k, \hat{g}_2 = 0, \hat{g}_{22} = k \sin^2\theta$

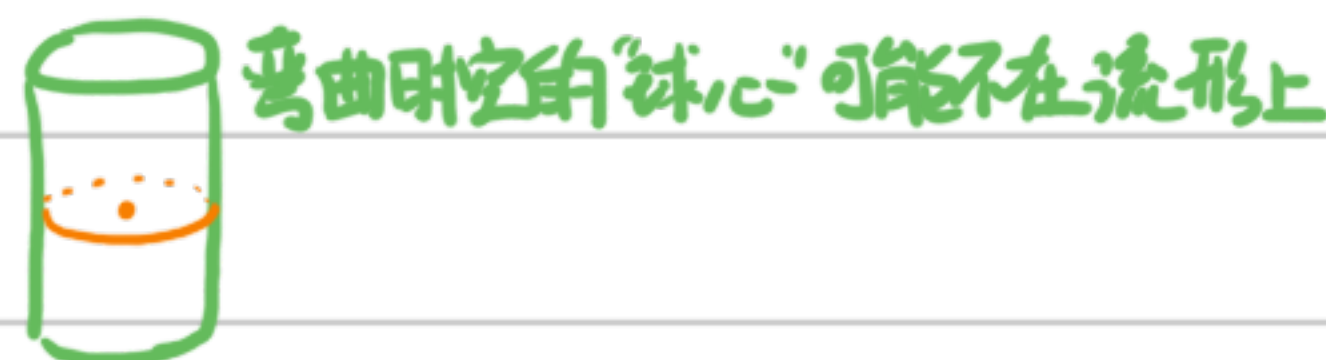
(M, g_{ab}) 的等度规子群 G_3 与 $SO(3)$ 同构

$SO(3)$ 群元 $\rightarrow$ 轨道球面: G_3 的所有轨道都为轨道球面

$d\hat{s}^2 = K(d\theta^2 + \sin^2\theta d\varphi^2)$ 为该度规 $\hat{g}_{ab}$ 的线元 $\rightarrow$ 面元 $\hat{e} = \sqrt{\hat{g}} d\theta \wedge d\varphi$ $A = \int_S \hat{e} = 4\pi K \rightarrow$ 定义 $r := \sqrt{A/4\pi}$ 则

$ds^2 = -dt^2 + dr^2 + d\hat{s}^2$

这里先取球面, 随后导出 r , 防止其在其它度规不成立
 故从对称性入手建系



3. 史瓦西真空解

静态球对称:

Claim: 静态球对称 (M, g_{ab}) 为仅有一个类时 Killing 场与面正交 (ξ^a), G_3 为等度规群中与 $SO(3)$ 同构的子群
 则: G_3 的所有轨道球面 $\perp \xi^a$ 中保度规, 使 $\xi^a \cdot T_a = 0$, 又 ξ^a 准-

Pf: $\phi \in G_3: M \rightarrow M$. 对 Killing ξ^a . $\forall \phi, \phi_* \xi^a = \xi^a$, (若不正交, 有分量 $\hat{\xi}^a$, $\phi_* \hat{\xi}^a \neq 0 \rightarrow$ 不成立)
 dt 与 $d\hat{s}$ 正交 ($d\hat{s}$ 切于轨道球面)

$$ds^2 = g_{00}dt^2 + g_{11}dr^2 + d\hat{s}^2 \rightarrow \text{对称性, } g_{00} \text{ 与 } g_{11} \text{ 与 } \theta, \varphi \text{ 无关}$$

$$\text{设 } d\hat{s}^2 = -e^{2A(r)}dt^2 + e^{2B(r)}dr^2 + d\hat{s}^2$$

解 $R_{ab} = 0$: 计算 $\Gamma^{\rho}_{\mu\nu}$: $g_{\mu\nu} = \begin{bmatrix} -e^{2A(r)} & & & \\ & e^{2B(r)} & & \\ & & r^2 & \\ & & & r^2 \sin^2\theta \end{bmatrix}$

$$R_{\mu\nu}{}^{\rho} = \Gamma^{\rho}_{\mu\sigma, \nu} - \Gamma^{\rho}_{\nu\sigma, \mu} + \Gamma^{\lambda}_{\sigma\mu} \Gamma^{\rho}_{\nu\lambda} - \Gamma^{\lambda}_{\sigma\nu} \Gamma^{\rho}_{\mu\lambda}$$

$$\begin{cases} \Gamma^{\rho}_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} (g_{\sigma\mu, \nu} + g_{\sigma\nu, \mu} - g_{\mu\nu, \sigma}) \\ \Gamma^0_{01} = \Gamma^0_{10} = A' ; \Gamma^1_{11} = B' ; \Gamma^0_{02} = A' e^{A(r)-B(r)} \\ \Gamma^1_{22} = -r e^{2B}, \Gamma^1_{33} = -r \sin^2\theta e^{2B} ; \Gamma^2_{21} = \Gamma^2_{12} = \frac{1}{r} = \Gamma^3_{13} = \Gamma^3_{31} \\ \Gamma^2_{33} = \frac{\sin 2\theta}{2} ; \Gamma^3_{23} = \Gamma^3_{32} = \cot\theta \end{cases}$$

$$\begin{aligned} R_{\mu 0} = R_{\mu\nu}{}^{\nu} : R_{00} &= -e^{2(A-B)} (-A'' + A'B' - A'^2 - 2\frac{A'}{r}) = 0 \\ R_{11} &= -A'' + A'B' + 2\frac{B'}{r} = 0 \\ R_{22} &= -e^{2B} (1 + r(A'-B')) + 1 = 0 \\ R_{33} &= R_{22} \cdot \sin^2\theta \end{aligned}$$

$$\begin{aligned} A' + B' &= 0 \rightarrow A = -B + \alpha \\ e^{2B} &= (1 + \frac{C}{r})^{-1} \end{aligned}$$

α 为常数, 使 ∂_t 仍为 Killing 场

$$\xrightarrow{ds} ds^2 = -(1 + \frac{C}{r}) e^{2A} dt^2 + (1 + \frac{C}{r})^{-1} dr^2 + d\hat{s}^2$$

$$ds^2 = -(1 + \frac{C}{r}) d\hat{t}^2 + (1 + \frac{C}{r})^{-1} dr^2 + d\hat{s}^2$$

$$\text{史瓦西度规 } \eta_{ab} + \gamma_{ab}, \gamma_{\mu\nu} = \begin{bmatrix} -\frac{C}{r} & & & \\ & \frac{C}{r} & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

$r \rightarrow \infty$ $ds^2 \rightarrow -d\hat{t}^2 + dr^2 + d\hat{s}^2$ 为闵氏

$$\frac{C}{r} \ll 1 : \gamma_{\mu\nu} \approx \begin{bmatrix} \frac{C}{r} & & & \\ & -\frac{C}{r} & & \\ & & 0 & \\ & & & 0 \end{bmatrix} \xrightarrow{\text{牛顿近似}} -\frac{C}{r} = 2\phi : C = -2M$$

$G=1, c=1$



$$P_1 - P_2 : L = \int (h_{ij} dx^i dx^j)^{\frac{1}{2}} = \int_{r_1}^{r_2} (1 - \frac{M}{r})^{\frac{1}{2}} dr > r_2 - r_1 \rightarrow \text{固有距离}$$

Birkhoff 定理: 真空球对称解必静态. 证

↓
球分布物质解均为史瓦西
↓
无单极辐射与偶极辐射

4. Vaidya度规

$$\text{史瓦西: } ds^2 = (1 - \frac{2M}{r}) dt^2 + (1 - \frac{2M}{r})^{-1} dr^2 + d\hat{s}^2 \quad (r > 2M)$$

$$\downarrow$$

$$r_* \equiv r + 2M \ln(\frac{r}{2M} - 1) \quad u \equiv t - r_* \quad \{u, r, \theta, \varphi\} \rightarrow ds^2 = -(1 - \frac{2M}{r}) du^2 - 2 du dr + d\hat{s}^2 = [-du^2 - du dr + d\hat{s}^2] + \frac{2M}{r} du^2$$

乌龟坐标 平直度规

$$\text{Vaidya度规: } ds^2(\text{vai}) = ds^2(\text{史}) + \frac{2m(u)}{r} du^2 \quad g: g_{uu} = -(1 - \frac{2m(u)}{r}), g_{ur} = g_{ru} = -1, g_{\theta\theta} = r^2, g_{\varphi\varphi} = r^2 \sin^2 \theta$$

$$g_{ab} = \sum g_{\mu\nu} (e^\mu)_a (e^\nu)_b = g_{uu} \cdot (du)_a (du)_b - (du)_a (dr)_b + r^2 d\theta^2_{ab} + r^2 \sin^2 \theta d\varphi^2_{ab}$$

$$g^{ab} = \frac{1}{r^2} \delta^{ab} + \frac{1}{r^2 \sin^2 \theta} \delta^{ab} - \frac{g_{ur}}{g_{uu}} (\frac{\partial}{\partial r})^a (\frac{\partial}{\partial u})^b + g_{uu} (\frac{\partial}{\partial r})^a (\frac{\partial}{\partial r})^b$$

$$\downarrow$$

$$\text{求}\Gamma: \Gamma^u_{uu} = -\frac{\dot{m}}{r^2} \quad \Gamma^u_{\theta\theta} = r \quad \Gamma^u_{\varphi\varphi} = r \sin^2 \theta$$

$$\Gamma^r_{uu} = \frac{(1 - \frac{2m}{r}) \dot{m}}{r^2} - \frac{\dot{m}}{r}, \quad \Gamma^r_{ur} = \Gamma^r_{ru} = \frac{\dot{m}}{r^2}, \quad \Gamma^r_{\theta\theta} = 2m - r, \quad \Gamma^r_{\varphi\varphi} = \Gamma^r_{\theta\theta} \cdot \sin^2 \theta$$

$$\Gamma^\theta_{r\theta} = \Gamma^\theta_{\theta r} = \frac{1}{r}, \quad \Gamma^\theta_{\varphi\varphi} = -\frac{\sin^2 \theta}{2}$$

$$\Gamma^\varphi_{r\varphi} = \Gamma^\varphi_{\varphi r} = \frac{1}{r}, \quad \Gamma^\varphi_{\theta\varphi} = \cot \theta$$

$$R_{uu} = \frac{-2\dot{m}}{r^2} \quad R = g^{uu} R_{uu} = 0$$

$$\text{爱因斯坦方程: } R_{ab} = 8\lambda T_{ab} \rightarrow T_{ab} = \frac{-\dot{m}}{4\pi r^2} (du)_a (du)_b \quad \text{能量辐射也可}$$

$$\text{令 } k_a \equiv (du)_a, \quad k^a = g^{ab} (du)_b = (\partial/\partial r), \quad T_{ab} = \frac{-\dot{m}}{4\pi r^2} k_a k_b, \quad \dot{m} < 0 \text{ 时 } T_{ab} = \Phi^2 k_a k_b$$

故 Vaidya 度规描述辐射场与无源类电磁场

$$\downarrow$$

无源类电磁场满足, $F^{ab} F_{ab} = 0$

$$\Phi^2 = \frac{E^2}{2\pi}$$

Kinnersley度规 (Vaidya度规推广)